Organisation X endpoint forensics — September 2026

Endpoint Compromise Forensic Analysis

of Multi-Endpoint Phishing & C2 Beaconing Campaign

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EXECUTIVE RISK OVERVIEW

Severity	Vector	Detail
HIGH	Persistent C2	Five hosts beacon 1–3 weeks post-quarantine via per-victim C2 domains paired with anonymized IPs in same-second sessions.
HIGH	Remote-access persist	Anydesk relays run after payload quarantine (EPGSQVEB); one phishing host never quarantined (EPLFOXSS, AV file-not-found 01–19 Sep).
MEDIUM	Trojanized installers	HP/printer bundles, freshysearch redirect chain, and XDG desktop persistence.
INFO	USB ruled out	EPD24DBC Mass Storage (22-09, post-window) uncorroborated; mouse benign. Phishing/drive-by, not a USB worm.

Scope	endpoint_security.csv (23,540 rows, 247 endpoints, 19 units) + usb_events.csv (8) + web_activity.csv (2,812,118 rows, 248 endpoints), all local UTC+05:30
Findings	33 endpoints FileType=Malicious (134 rows, 60 dedup) + 2 AV-only active + 8 C2 clusters

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1 Executive Summary

Endpoint telemetry across ˜250 Linux desktops (19 units, September 2026) characterises a multi-vector phishing and drive-by campaign staging pseudorandom and lure-themed payloads from user-writable paths. Artefact review confirmed quarantined payloads, a failed mail-payload remediation, per-victim C2 beaconing, hijacker bundling, and post-payload remote-access sessions. Verified file hashes, beacon indicators, and MITRE mappings are provided for endpoint containment.

Risk Assessment (for leadership)

Severity	Area	Detail
HIGH	Staging at scale	33 endpoints quarantined FileType=Malicious material (134 rows, 60 dedup endpoint+SHA), including a 9-host shared xbel payload.
HIGH	Beaconing persistence	Five hosts re-contact per-victim C2 for days to weeks after quarantine; Anydesk persists post-payload on one host.
MEDIUM	Installer trust abuse	Trojanized printer and HP bundles plus a browser-hijacker chain defeat signature-only inspection.
INFO	Detection gap	AV never logged a FOUND line (all summaries Infectedfiles:0); 179 action rows across 13 hosts carry the signal, and two AV-only hosts never reached quarantine.

Key Concepts (plain-English primer)

C2 beacon pair. The agent logs each contacted destination separately, so one connection appears as two same-second rows: a domain plus the IP it resolved to. Repeated domain+IP pairs at fixed times mean automated beaconing, not browsing.
Rare-domain cardinality. Counting how many endpoints ever contacted each destination separates campaign infrastructure (1–3 endpoints) from benign shared services (hundreds). LAN suffixes (`.hgu_lan`, `.bbrouter`, `.dlink`, `.local`) are stripped first.
Anonymized indicators. Challenge data replaces real C2 with `svc-<hex>.gov.example` domains and `198.18.0.0/15` IPs; widespread `svc-*` names are genuine gov portals and are excluded.
Clock skew. Endpoints upload days late with wrong clocks, so timelines use `event_time` with `received_time` as a sanity band (quarantine median ˜0h; av_scan median 34h).
Quarantine re-listing. The service re-lists the whole quarantine folder per scan, so one file appears many times; analysis dedups by endpoint+SHA (134 rows → 60).

2 Tools Used

Only tools that produced concrete evidence in this investigation are listed.

- **Python 3.12 + pandas (chunked)** — endpoint triage and 2.8M-row web cardinality/C2 scoring
- **rich** — terminal evidence tables (`rich_triage.py`)
- **uv** — reproducible environment (`uv sync`, `uv run`)
- **MiKTeX latexmk** — this report build (`Latex/main.tex` → PDF)

3 Methodology

3.1 Analysis

Quarantine FileType parsing (`FileType:` field with dedup), AV action-line review (`Beginning quarantine / Attempting / File not found`), web_block baselining, USB correlation, then chunked web_activity rare-domain scoring with same-second pair resolution and clock-skew banding.

3.2 Finding

Phishing and drive-by traits (lure documents, mail payload, hijacker bundling, trojanized installers), per-victim C2 pairs on 8 clusters, Anydesk remote access, `.desktop` persistence, and two AV-remediation gaps; USB and web_block ruled out as vectors; shared xbel hash links 9 hosts.

3.3 Outcome

Actionable IoCs (60 hashes, paths, 8 C2 clusters, MITRE map) compiled for isolation, firewall blocks, and EDR hunts.

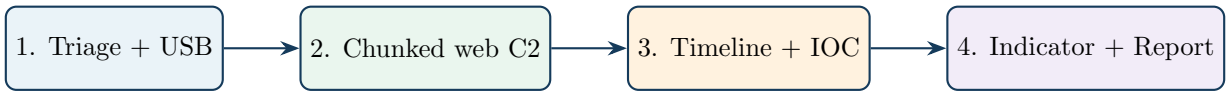
4 Reconstructed Timeline

Quarantine dates are first-seen `Quarantined At`; web sessions fall in the 10–21 Sep window. Isolate and re-image any host showing this sequence.

Timestamp	Event	Evidence
31-08 13:21	EPZRAPEL file quarantined (pre-window)	53EDA42869D4... hex-named file; C2 follows 11-19 Sep.
02-09	First xbel wave	EPMQTKSL + EPC2L3UH (18 re-lists, heaviest) share SHA 5e10b349....
03/07/08-09	Lure + installer wave	Dash Camera PPP pdf, RAGHAV PDFs, printer binaries, win-lbp exe, oscar xbel.
09/10-09	Juliett + kilo hits	EPFQSCDV xbel; EPEEHWSG Library Books.xlsx.
10-09 11:22	Hijacker chain starts	EPSES7ZK freshysearch 4-way redirect bursts begin.
11-09	Peak day	bundle.js x20, Final_Documents.zip, Dash Camera pdf, EPVC2P55 C2 15:58, EPZRAPEL C2 19:27.
12-09	Mail + staging C2	EPLFOXSS pairs; EPWZ7L6L staging; shared n7093/n3618 cluster active.
13/14-09	Document wave	mydoc x4, RETIREES WELFARE xlsx; EPVC2P55 bursts.
15-09	Anydesk + cab + banking	EPGSQVEB sessions; Data1.cab 11:28; EPVSWBNA banking (pre-infection).
17-22-09	Late hits + USB	Lakeside docs, EPNPX6VY dll, HP x5 AV detect; EPD24DBC USB 22-09 (post-window).
23-26-09	Tail quarantines	Printer DLLs AV, libreoffice .desktop x2, dash cam pdf, xlsx/docx/so (post-window).

5 Step by Step Process Along with Screenshots

Architecture Overview



Step 1: Endpoint Triage & Cryptographic Baselining

Parsed 23,540 rows: 134 FileType=Malicious rows across 33 endpoints (60 dedup), 0 FOUND lines, 179 AV action rows, 914 policy web blocks, 8 USB rows with zero compromised overlap:

```
uv run python rich_triage.py
# scope, quarantine, AV gaps, USB, skew, C2 spot counts
```

Step 2: Chunked Web C2 Hunt

Rare-domain scoring over 2,812,118 rows (19,027 uniq destinations) isolated per-victim beacons; same-second domain+IP pairs confirm sessions. Widespread gov/Mozilla/Yahoo infra excluded as benign.

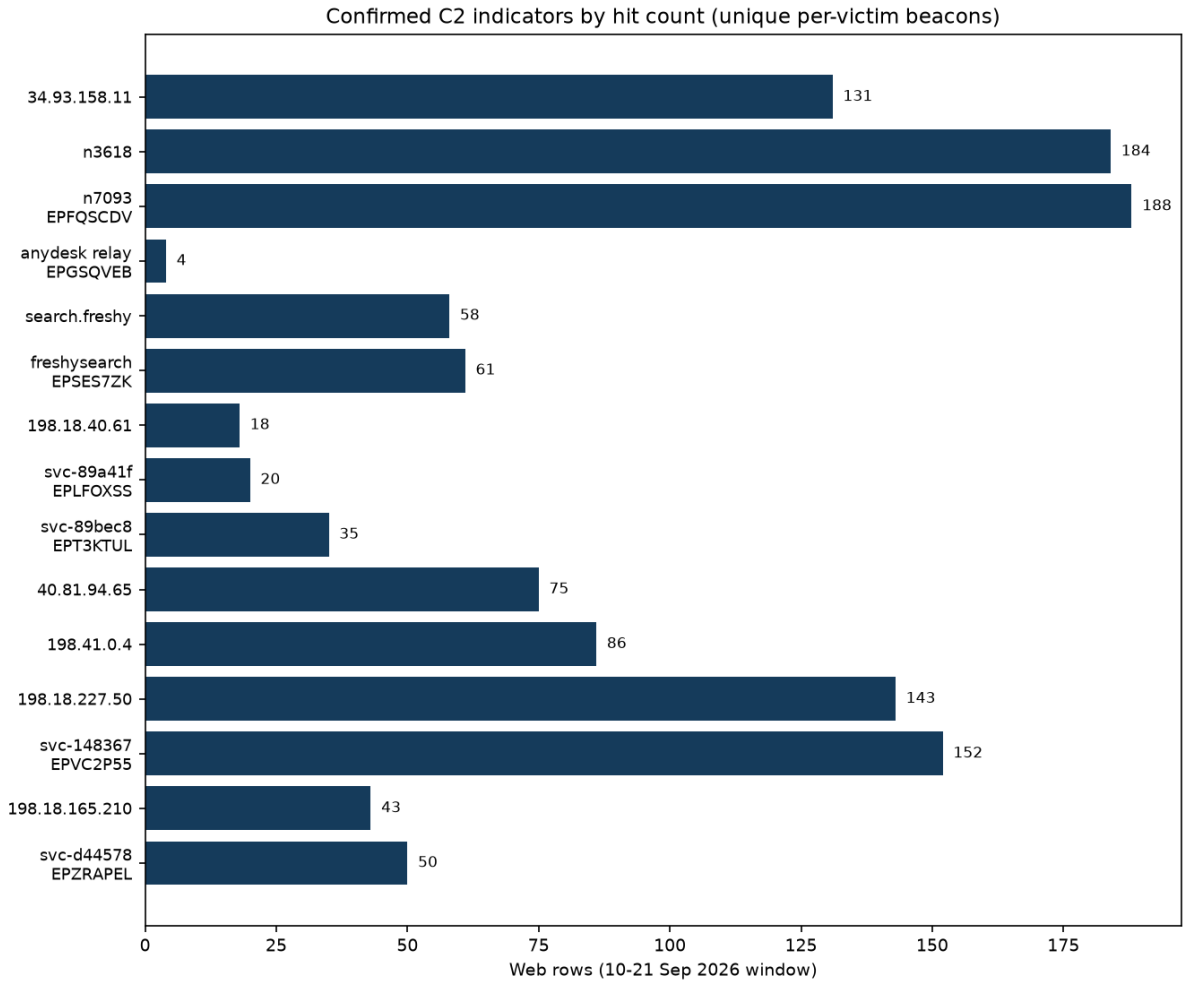


Figure 1: Confirmed C2 indicators by hit count (verified web counts).

Step 3: Timeline & Clock-Skew Review

Per-endpoint session dates (UTC+05:30) with quarantine median ˜0h and av_scan median 34h bands; late uploaders (EPC2L3UH 18.4h, EPLFOXSS 31.8h, EPGSQVEB 32.9h) flagged offline. Terminal evidence tables are produced by rich_triage.py; save a screenshot as im-

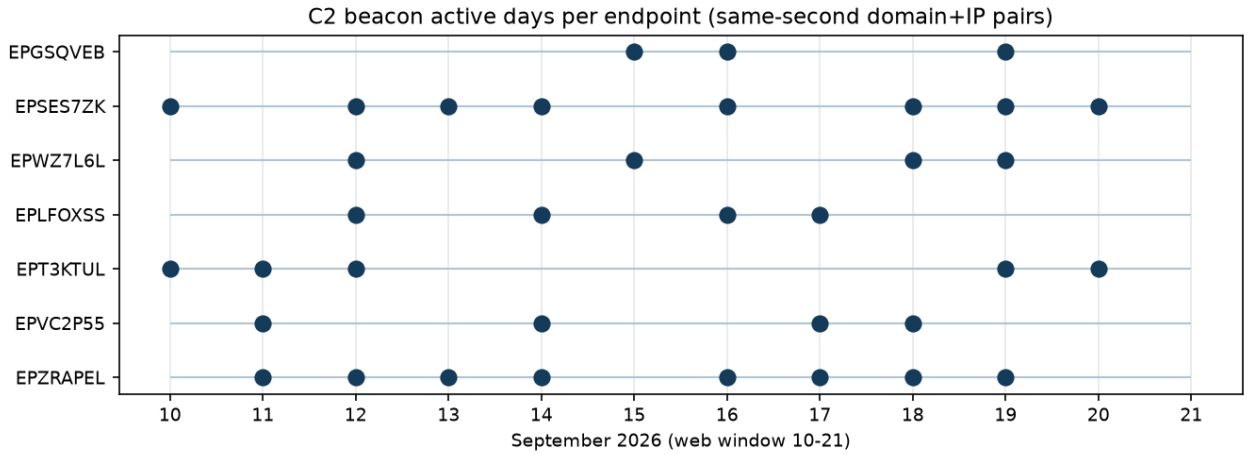


Figure 2: C2 beacon active days per endpoint.

ages/triage_tables.png to embed it here.

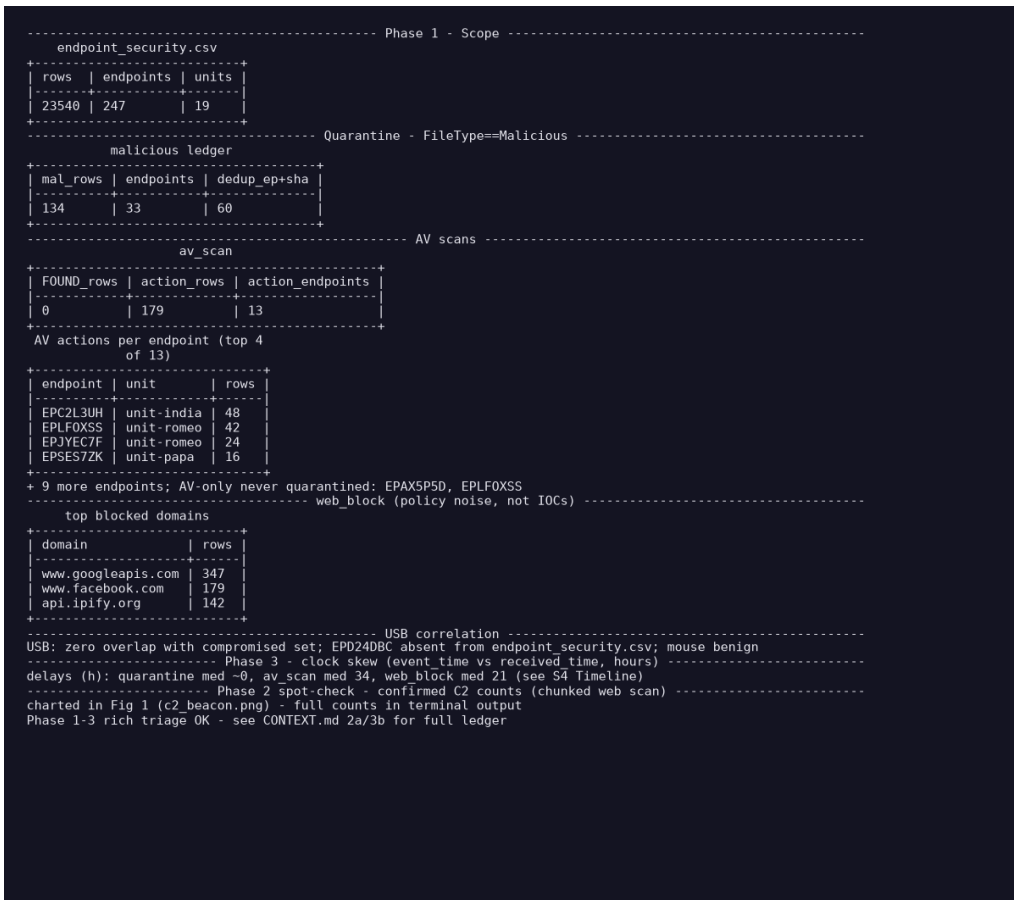


Figure 3: Rich Phase 1 terminal tables (abridged; full output via rich_triage.py).

Step 4: Indicator Extraction & Reporting

```
uv run python phase1_triage.py  # output/phase1_iocs.csv (60 rows)
uv run python gen_figures.py  # Latex/images/c2_beacon.png, timeline.png
uv run python generate_latex.py  # Latex/main.tex
cd Latex && latexmk -pdf main.tex
```

6 Technical Findings Summary

Findings below use wrapped monospace cells so hashes and paths never overflow margins.

Endpoint	Payload	C2 + counts	Persistence
EPZRAPEL	`53EDA42869D4... (hex-named)`	`svc-d44578 (50) + 198.18.165.210 (43); 11-19 Sep pairs`	Persists 3 wks post-quarantine
EPVC2P55	`Data1.cab`	`svc-148367 (152) + 198.18.227.50 + 198.41.0.4 + 40.81.94.65`	C2 after 15-09 quarantine
EPT3KTUL	`Dash Camera-49600.00.pdf`	`svc-89bec8 (35) + 198.18.251.94 + n7473 sub`	Persists to 20-09
EPLFOXSS	`Evolution-mail payload (unquarantined)`	`svc-89a41f (20) + 198.18.40.61 (18)`	AV File-not-found 01-19 Sep
EPWZ7L6L	`ops-electron-app.desktop`	`svc-b76574 (13) + svc-171e30 (10) + n2b40`	Staged pre-25-09 file
EPSES7ZK	`hp_LJ1020 x5 (same SHA)`	`freshysearch + search.freshy + searchtabnew + cloudfront`	Hijacker + installer
EPGSQVEB	`Final_Documents.zip`	`Anydesk relays + 148.113.17.93/.8.189`	Sessions post-payload
EPFQSCDV	`recently-used.xbel`	`n7093 (188) + n3618 (184) + 34.93.158.11; shared eps3`	2 co-suspects triage
Others (25)	`xbel / lure docs / printer / desktop / cab / dll`	`No unique beacon in window`	File evidence; cleaned or out-of-window

7 MITRE ATT&CK Mapping

Technique	Name	Observed Behaviour
T1566.001	Spearphishing Attachment	Lure PDFs/DOCX/XLSX (Dash Camera, RAGHAV, Lakeside, RETIREES, Staff Library) and the evolution-mail payload on EPLFOXSS.
T1204.002	Malicious File	User execution of quarantined payloads: documents, HP/printer installers, .desktop launchers, Data1.cab, Anydesk-linked zip.
T1036	Masquerading	Payloads reuse legitimate names and paths: recently-used.xbel, libreoffice-calc.desktop, printer DLL names, ops-electron-app.desktop.
T1071.001	Web Protocols	Per-victim C2 beacons (svc-¡hex¿.gov.example + 198.18.x.x same-second pairs) and the freshysearch redirect chain.
T1219	Remote Access Tools	Anydesk relay sessions on EPGSQVEB on 15/16/19 Sep, all after the 11-09 payload quarantine.
T1574.002	DLL Side-Loading (suspected)	Trojanized printer bundle layout on EPVSWBNA (launcher.exe with Strings.dll / zshp1020s.dll); unconfirmed at dynamic level.

8 Recommendations

1. Isolate still-compromised hosts (EPZRAPEL, EPVC2P55, EPT3KTUL, EPLFOXSS, EPWZ7L6L, EPSES7ZK, EPGSQVEB, EPFQSCDV); triage co-suspects EPCRFN4D/EPV7MB2K/EPL4KXNW; collect quarantine dirs and agent logs before re-imaging.

2. Hunt retrospectively for the shared xbel SHA, the HP/printer bundle SHAs, and per-victim svc beacons using the verified hashes in Appendix A.

3. Remove desktop-file persistence and audit home download/desktop/share folders for residual payloads; revoke Anydesk access on EPGSQVEB.

4. Future work: run an authorised isolated capture to confirm callback behaviour of the anonymized C2 endpoints; do not assert payload contents beyond current evidence.

A Appendix A: Pipeline-Extracted IOCs (60 dedup rows)

Endpoint	File	SHA-256
EP2SJ6NE	libreoffice-calc.desktop	93ffd792f92437a850e4939c84442391a624e20446cea276fffcfa2a4c664548
EP3GALVS	recently-used.xbel	5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c
EP4YIFHV	load-hub-i18n.bundle.js	036965c24ac54284e0bfac480395c91214e7b3c64d00215266a75e8977140f10
EP4YIFHV	shopping_iframe_driver.js	100d970fc784e9bcb37b707ec4ef69825b165a073659f4068070a165184a087f
EP4YIFHV	tokenized-card.bundle.js	39f5d83d7982cfae057053b0dcffa66fca12bee9769b1587a27e4387f56f29ec
EP4YIFHV	wallet_checkout_autofill_driver.js	444d4db051edabd5abb29d93dcc5b07e0c662b03e2d785e4a28dfb7fc8032fc4
EP4YIFHV	wallet-buynow.bundle.js	4aab5c4ac7a5cf0aab311362514cda20a2f3beb777a553756bf8ad8c7dfaa2bd
EP4YIFHV	wallet_donation_driver.js	528c866fabbe17c30e9790a12ac09b94dd820ba4e46b0f9d9bb78a2795992f1c
EP4YIFHV	runtime.bundle.js	569e425e7f048b1f769c22d65aa91427b69c3d248068af5e628cd9467ef250ca
EP4YIFHV	load-ec-i18n.bundle.js	56bada50510d97878d514009279c3a662d9894a21c344fea8f9f9c5e3094614f
EP4YIFHV	bnpl.bundle.js	620389d26167bea08e56a90480f74d3dbd32c3e974f4de7a950953663a3fe6a1
EP4YIFHV	bnpl_driver.js	6c21396c9518fa414ddb4dada2adb09e2d2b2fc5069d89da7e9df2adf2d95287
EP4YIFHV	miniwallet.bundle.js	6f47d71e9ee52102a57e892441333ce18f60ba3e356c565b4742ff12e465234a
EP4YIFHV	notification_fast.bundle.js	876ede3dba963ec8e7b99252979f3825807187785abb3759b99e3d35e30693fd
EP4YIFHV	crypto.bundle.js	8e661de94430dc0293ca7964f0f9b89682e8fd2f56c17fb4b3c3de242d4f58d0
EP4YIFHV	edge_driver.js	985d0862ba672e36839438e44cbaf6d18537b53010b778a94bd7d7f2fa85af13

Endpoint	File	SHA-256
EP4YIFHV	notification.bundle.js	b5aa710d917ed708f370fe187bbc5fc452e3bb66af0ed4f642e54cd155e86524
EP4YIFHV	buynow_driver.js	bee346d14d293f9f4b989362041875b0257477a77759855e41c2f3cd40a6bed2
EP4YIFHV	c3f9c4a416e9e493005fc3f97adacbcf12956a81cc6c645d255fd023442c8120	c3f9c4a416e9e493005fc3f97adacbcf12956a81cc6c645d255fd023442c8120
EP4YIFHV	app-setup.js	d604ac767e070cf98b235b0281ef932c463fa11bceaf2fbf743bbf33743b0af1
EP4YIFHV	webui-setup.js	e3d95016d065059105580ae8ba811cf4bb130b515b8c77e5e58cf95763ad938a
EP4YIFHV	load-ec-deps.bundle.js	f2bf148b4d999cb41ec764960ee718e0c9986a749f3b7be372ad86cf98f323de
EP7UYCXM	libeToken.so.10.7.77	574564cd6ed90f7938ab4967d5802f28cddc38ed1148e42838083ea6553c3be5
EPAGTKNF	Buy CP PLUS Dashboard Camera, Warranty NONE 1 year online _ Public Procurement Portal (PPP).pdf	18866b6f7308f73bc1b63193e50629bfd4fff701d6b9d3d2d6331b90e9162208
EPAJI3E5	Lakeside traders.docx	c11e86d9e30b5fad8daf1c15c4dbbfd8d31b581a5d930b9c66d996d3bda11a2f
EPAJI3E5	Lakeside traders.pdf	da22b024b1ba07c367e812e3c2f889f60cd3eb9758be906f6d5c231b8b71937b
EPC2L3UH	recently-used.xbel	5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c
EPDRB236	recently-used.xbel	5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c
EPEEHWSG	Library Books.xlsx	98d119993c27068f00bdf18a17d6613ae6ce6046b6b6620979282c2662c4efbd
EPF4AJCM	libreoffice-calc.desktop	93ffd792f92437a850e4939c84442391a624e20446cea276fffcfa2a4c664548
EPFQSCDV	recently-used.xbel	5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c
EPGSQVEB	Final_Documents.zip	404ee57531c33156d62154f49e3700c8c1f09cd0c75692f795748d59b9145088
EPJ42VCZ	RAGHAV STNA STNB.pdf	854b8f7716627df333e8ecfeae5887c02b3195658f7394573fe21b67e93e11be
EPJ42VCZ	RAGHAV STNC STNA.pdf	a2beca1870ae689f4b046a20c3d8e2e48557d782dea53489b21a8b0adcef38be
EPJ76RZV	COMPRESSED FINAL COMPARISION RETIREES WELFARE1.xlsx	d4a23078663777e758c04d2670b0d6aac8aa7ba5f91f811c5d94fa6ce8b7ef56

Endpoint	File	SHA-256
EPJYEC7F	recently-used.xbel	5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c
EPLDM6PE	recently-used.xbel	5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c
EPMDU6L5	recently-used.xbel	5dd59e8309b4be90aad58871d82a5b3fb6dcd7ff19a7fb34587732512b28739d
EPMQTKSL	recently-used.xbel	5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c
EPMVZYZO	mydoc-10.pdf	5b7914202b30fc7e28ddcd2f82502aa8531035b812ee2194d978b7e920f43066
EPMVZYZO	mydoc-9.pdf	7c10f54076fba8141f8ef95a701d0a4bdadfc63a5b98bec16b6b68439ebc3dc7
EPMVZYZO	mydoc-75.pdf	92e24303edb5f2b6f1bc6c0c375655717eb2227c1622c4a723791aafce33965d
EPMVZYZO	mydoc.pdf	d474833503a6404b8106d981c9f0769e4bd97033f45cf424e360b61f659c8cb1
EPNPX6VY	Microsoft.Practices.EnterpriseLibrary.Common.dll	bde5bba12f076659ab19d3df625dedc7211180038c821b77b431b62647ef231f
EPOC6Y4R	recently-used.xbel	5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c
EPPGWIOF	rawtobr3	64efda7704564dd50c7514e74ef3c5252bc68af8f3b0d92358b64771980665af
EPPGWIOF	brprintconflsr3	6594cac08d03782ad43d6b293faff904f20032c8eb37a1b27b6caf57b1aeae5e
EPQBJBDX	dash cam.pdf	30c78bc9ec28a6e4d47708c309f8c88599c71a0fb379a3b4d75847d0f66ef0ca
EPQKOBYG	win-lbp623-621-fw-v1301(1).exe	f475a7462dd85932ad7f99354751663d1543adb5e37c7c51ece78e10b6c37189
EPR4BEDS	recently-used.xbel	5e10b3490160cca0fc872f2b026d4b40bafc236852f83d30c9031899e50adc9c
EPRD7FJT	12th Plan Works Dashboard.pdf	5c7fc859c5a8f2bba02446c30020bb2b478686d66b566818c7382c840d95720f
EPSES7ZK	hp_LJ1020_Full_Solution-v2012_918_1_57980(1).exe	f0c485571570100898f2d15531fbe4531ba303a252cd40a79b64c1976a5a9890
EPSJWX5J	Staff_Library_Payment_Final.docx	d831cf990c1445b507ef9fea05a6e483983aa4737fa56fcf650a96af1cee0cd5
EPT3KTUL	Dash Camera-49600.00.pdf	7505de4759aadebf5e21b2cc0f7f13fccdbec968e3b82be70791197fac3d21bc

Endpoint	File	SHA-256
EPVC2P55	Data1.cab	925b63ce20d94bc4a0d98d02 d153161dbba9ac61c39ec323 ac8a73b2acbde42e
EPVSWBNA	launcher.exe	61a67e53c6c0c80a4e95af35 a2997ad2ce59720020fc7438 991eda3541df73b6
EPVSWBNA	Strings.dll	ae107170c153cb3eb34c0494 82722d3937f2c03a6c23cadb 9bbdb9034e28b9c0
EPVSWBNA	zshp1020s.dll	c6525c5230387653e16f21f4 08eab8202157b0eddeb180bd 91a13e860fbee4f4
EPWZ7L6L	ops-electron-app.desktop	dd294df9886197bd68807cee 3bf62d10859b4860bd95421d da6bb67db9648dd1
EPZRAPEL	53EDA42869D426262B752F088EA605187284ACE8	2e4dff1672bbabba74a0a99d 54e38e56e036ce8471989583 039b3c3e7edac0b7

B Appendix B: Reproducibility

```
uv sync
uv run python rich_triage.py
uv run python phase1_triage.py
uv run python gen_figures.py
uv run python generate_latex.py
cd Latex && latexmk -pdf main.tex
```